The Integers and the Real Numbers Solutions

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September 13, 2026

Exercises

1. Prove the following "laws of algebra" for $\mathbb{R}$, using only axioms (1)-(5):

(a) If $x + y = x$, then $y = 0$.
Proof. By axiom (3) we have that there exists a unique element of $\mathbb{R}$ called zero denoted by 0, such that $x + 0 = x$ for all $x \in \mathbb{R}$. Given that $x + y = x$ and that 0 is unique, we conclude that $y = 0$ must be true. □

(b) $0 \cdot x = 0$. [Hint: Compute $(x + 0) \cdot x$.]
Proof. Consider the product $(x+0)\cdot x$. By axiom (5) we have that this product is equivalent to $x \cdot x + 0 \cdot x$. To show that the identity we aim to prove holds we need only show that $x \cdot x + 0 \cdot x = x \cdot x$.

$$x \cdot x + 0 \cdot x = (x + 0) \cdot x + 0 \cdot x$$

$$x \cdot x + 0 \cdot x = x \cdot x + 0 \cdot x + 0 \cdot x$$

$$x \cdot x = x \cdot x + 0 \cdot x$$

Thus, by part (a) we conclude that $0 \cdot x$ must be 0. □

(c) $-0 = 0$.
Proof. Recall that due to notation -0 is the element in $\mathbb{R}$ such that $0 + (-0) = 0$. Thus, by (a), we conclude that $-0 = 0$. □

(d) $-(-x) = x$.
Proof. Due to notation, note that $-x$ is the number in $\mathbb{R}$ such that $x + (-x) = 0$. Furthermore, we also have that $(-x) + (-(-x)) = 0$. By axiom (4) the negative of a number in $\mathbb{R}$ is unique. Thus, it must be that $x = -(-x)$. □

(e) $x(-y) = -(xy) = (-x)y$.
Proof. To prove the first equality, $x(-y) = -(xy)$, consider the following expression: $xy + x(-y)$. By axiom (5), we have that the expression is equal to $x(y + (-y))$. Similarly, note that $y + (-y) = 0$ by axiom (4), showing that $xy + x(-y) = x \cdot 0$. By part (b), we conclude that $xy + x(-y) = 0$. Therefore, it must be that $x(-y) = -(xy)$.
For the second equality, $-(xy) = (-x)y$, we follow an analogous argument by looking at the expression $xy + (-x)y$ and note that it is equivalent to 0. Thus, we conclude that $-(xy) = (-x)y$. □

(f) $(-1)x = -x$.
Proof. By part (e) we have that $(-1)x = -(1x)$. By axiom (3), we have that $1 \cdot x = x$. Thus, we have that $(-1)x = -(1x) = -x$. □

(g) $x(y - z) = xy - xz$.

Proof. Note that $x(y-z) = x(y+(-z))$. Thus, by axiom (5), $x(y-z) = xy + x(-z)$. In part (e) we proved that $x(-z) = -(xz)$ and thus, $x(y-z) = xy - xz$. □

(h) $-(x+y) = -x - y; -(x-y) = -x + y$.
Proof. Note $-(x+y) = -1 \cdot (x+y)$. Hence, by axiom (5), $-(x+y) = (-1)x + (-1)y$. In part (f) we proved $(-1)x = x$, thus we conclude $-(x+y) = -x - y$.
Similarly, $-(x-y) = -1 \cdot (x+(-y))$, which analogously expands as $(-1)x + (-1)(-y)$. By part (f) $(-1)x = -x$ and $(-1)(-y) = -(-y)$. Furthermore, in part (d) we proved $-(-y) = y$. Thus, we conclude $-(x-y) = -x + y$. □

(i) If $x \neq 0$ and $x \cdot y = x$, then $y = 1$.
Proof. Since $x \neq 0$, by axiom (4) we have that $1/x$ exists such that $x(1/x) = 1$. Thus, consider the expression $(1/x)(xy)$ which is equivalent to $(1/x)x$ since $xy = x$. By axiom (1) we have that $(1/x)(xy) = (1/x \cdot x)y = 1 \cdot y = y$. Since this expression is equivalent to $1/x \cdot x = 1$, we conclude that $y = 1$. □

(j) $x/x = 1$ if $x \neq 0$.
Proof. Let $x \neq 0$ and recall the notation that $1/x$ is the unique element from axiom (4) such that $x \cdot (1/x) = 1$. Furthermore, that x/x is another notation for $x \cdot (1/x)$. Therefore, it is a consequence of axiom (4) that $x/x = 1$ for $x \neq 0$. □

(k) $x/1 = x$.
Proof. Note the notation can be expressed as $x/1 = x(1/1)$ where $(1/1)$ is the unique element in $\mathbb{R}$ such that $1 \cdot (1/1) = 1$. By part (j) we conclude that $1/1 = 1$. Hence, $x/1 = x \cdot 1 = x$. □

(l) $x \neq 0$ and $y \neq 0$ implies $xy \neq 0$.
Proof. Let $x, y \neq 0$. Assume for contradiction that $xy = 0$. Recall in part (b) we proved that $0 \cdot x = 0$, thus $xy = 0 \cdot x$. As such, $(1/x)(xy) = (1/x \cdot x)y = y$ by axiom (1). Similarly, $(0 \cdot x) \cdot (1/x) = 0 \cdot (x \cdot 1/x) = 0 \cdot 1 = 0$. Thus, we conclude that $y = 0$, which is a contradiction to our initial condition on y. Therefore, $xy \neq 0$. □

(m) $(1/y)(1/z) = 1/(yz)$ if $y, z \neq 0$.
Proof. Let $y, z \neq 0$. Consider the product $yz \cdot [(1/y)(1/z)]$. By axiom (1) we have that this expression is equivalent to $z \cdot [y(1/y)] \cdot (1/z)$. By axiom (4) the expression is equal to $z \cdot (1/z) = 1$. Thus, by axiom (4) it must be that $(1/y)(1/z) = 1/(yz)$. □

(n) $(x/y)(w/z) = (xy)/)(yz)$ if $y, z \neq 0$.
Proof. Recall by notation that $x/y = x(1/y)$ and $w/z = w(1/z)$. Thus, we have $(x/y)(w/z) = [x(1/y)][w(1/z)]$. Using axiom (2) $[x(1/y)][w(1/z)] = [xw][(1/y)(1/z)]$. By part (m) $(1/y)(1/z) = 1/(yz)$. Hence, $(x/y)(w/z) = (xw)[1/(yz)] = (xw)/(yz)$. □

(o) $(x/y) + (w/z) = (xz + wy)/(yz)$ if $y, z \neq 0$.
Proof. Recall from notation that we can rewrite our initial expression as $x(1/y) + w(1/z)$. Note that $(yz)/(yz) = 1$, and thus

$$x(1/y) + w(1/z) = [(yz)/(yz)][x(1/y) + w(1/z)]$$

Using axiom (2) we have that

$$x(1/y) + w(1/z) = (yz)[x(1/y) + w(1/z)][1/(yz)]$$

By axiom (5)

$$x(1/y) + w(1/z) = [(yz)x(1/y) + (yz)w(1/z)][1/(yz)]$$

Using both axioms (1), (2), and (4)

$$x(1/y) + w(1/z) = [zx + yw][1/(yz)] = (xz + wy)/(yz)$$

□

(p) $x \neq 0$ implies $1/x \neq 0$.
Proof. Let $x \neq 0$. If $1/x = 0$, then by part (b) $x \cdot 1/x = 0$. But, by axiom (4) it must be that $x \cdot 1/x = 1$. Thus, it cannot be that $1/x = 0$. □

(q) $1/(w/z) = z/w$ if $w, z \neq 0$.
Proof. Let $z, w \neq 0$. First note that $1/(w/z) = 1/[z(1/w)]$. Thus, by part (m) we have $1/(w/z) = 1/w \cdot [1/(1/z)]$. Now, note by notation that $1/(1/z)$ is the element in $\mathbb{R}$ such that $(1/z) \cdot 1/(1/z) = 1$. In particular, by axiom (4) this element is unique. Thus, it must be that $z = 1/(1/z)$. Hence, $1/(w/z) = 1/w \cdot z = z/w$. □

(r) $(x/y)/(w/z) = (xz)/(yw)$ if $y, w, z \neq 0$.
Proof. Let $y, w, z \neq 0$. First note $(x/y)/(w/z) = (x/y) \cdot [1/(w/z)]$. Using part (q), $(x/y)/(w/z) = (x/y)(z/w)$. Finally, using part (n), $(x/y)/(w/z) = (xz)/(yw)$. □

(s) $(ax)/y = a(x/y)$ if $y \neq 0$.
Proof. Let $y \neq 0$. Note $(ax)/y = (ax) \cdot (1/y)$. Using axiom (1), this expression is equivalent to $a(x \cdot 1/y) = a(x/y)$. □

(t) $(-x)/y = x/(-y) = -(x/y)$ if $y \neq 0$.
Proof. Let $y \neq 0$. Note $(-x)/y = [-1 \cdot x] \cdot 1/y$. Using axiom (1) we have the equivalent expression $-1 \cdot (x \cdot 1/y) = -1 \cdot (x/y) = -(x/y)$ [We used part (f) for the final step]. Going back to the expression $-1 \cdot (x \cdot 1/y)$, using both axiom (2), (1), and part (f) we get the equivalent expression $x \cdot (-1 \cdot 1/y) = x \cdot [-(1/y)]$. In particular, note $-(1/y)$ is the element in $\mathbb{R}$ such that $-y \cdot -(1/y) = 1$. Thus, $-(1/y) = 1/(-y)$ by axiom (4). Therefore, $x \cdot [-(1/y)] = x \cdot 1/(-y) = x/(-y)$. □

2. Prove the following "laws of inequalities" for $\mathbb{R}$, using axioms (1)-(6) along with the results from **1.**

(a) $x > y$ and $w > z \Rightarrow x + w > y + z$.
Proof. Let $x > y$ and $w > z$. By axiom (6), $x + w > y + w$ and $w + y > z + y$. Therefore, $x + w > z + y$. □

(b) $x > 0$ and $y > 0 \Rightarrow x + y > 0$ and $x \cdot y > 0$.
Proof. Let $x > 0$ and $y > 0$. To prove $x + y > 0$, note that by part (a) we have $x + y > 0 + y = y$. Since $y > 0$, $x + y > 0$.
To prove $x \cdot y > 0$, we make use of axiom (6). Since $x > 0$ and $y > 0$, by axiom (6) $x \cdot y > 0 \cdot y$. Using part (b) from **Exercise 1**, $x \cdot y > 0$. □

(c) $x > 0 \Leftrightarrow -x < 0$.

Proof. Let $x > 0$. Then, by axiom (6), $x + (-x) > 0 + (-x)$. This expression is equivalent to $0 > -x$ by axioms (3) and (4).
Let $0 > -x$. Then, by axiom (6) $0 + x > -x + x$, which is equivalent to the expression $x > 0$ by both axioms (3) and (4). □

(d) $x > y \Leftrightarrow -x < -y$.
Proof. Let $x > y$. Consider the case $x = 0$, implying $y < 0$. By part (c), $-y > 0$ and since **Exercise 1 (c)** shows $0 = -0$ we have $-x = 0$. Thus, $-x < -y$.
Now consider the case $x < 0$. Since $y < x$, it must be $y < 0$. By part (c), $-x > 0$ and $-y > 0$. Using axiom (6), we note $x + (-x) > y + (-x)$, which is equivalent to the expression $0 > y + (-x)$. Using axiom (6) once more we obtain the equivalent expression $0 + (-y) > y + (-x) + (-y)$. Simplifying, we obtain the equivalent expression $-y > -x$.
Finally, we consider the case $x > 0$. If $y = 0$, we find a case analogous to when $x = 0$, under which by following it's logic we once again land that $-y > -x$. If $y < 0$, we note that $-y > 0$ and that $-x < 0$, and thus $-y > -x$. If $y > 0$, we make use of axiom (6) to obtain the expression $x - x > y - x$, which is equivalent to $0 > y - x$. Using axiom (6) once more we obtain the expression $0 - y > y - x - y$, which is equivalent to $-y > -x$. □

(e) $x > y$ and $z < 0 \Rightarrow xz < yz$.
Proof. Let $x > y$ and $z < 0$. By part (c) $-z > 0$. Using axiom (6) we have $x(-z) > y(-z)$. By part (e) in **Exercise 1** we have the equivalent expression $-(xz) > -(yz)$. Finally, by part (d) $xz < yz$. □

(f) $x \neq 0 \Rightarrow x^2 > 0$, where $x^2 = x \cdot x$.
Proof. Let $x \neq 0$, and denote $x^2 = x \cdot x$. If $x > 0$, by axiom (6) we have that $x^2 > 0$.
Let $x < 0$. Then, by part (e) we note $0 \cdot x < x \cdot x$, which is equivalent to the expression $0 < x^2$. □

(g) $-1 < 0 < 1$.
Proof. We first note $1 \cdot 1 = 1$ by axiom (3). Thus, by part (f) it must be that $1 > 0$ since $1 = 1^2$. Since $1 > 0$, by part (c) it must be $-1 < 0$. Therefore, $-1 < 0 < 1$. □

(h) $xy > 0 \Leftrightarrow x$ and y are both positive or both negative.
Proof. Let $x > 0$ and $y > 0$. Then, by part (b) $xy > 0$. Let $x < 0$ and $y < 0$. Then, by part (c) $-x > 0$ and $-y > 0$. Thus, by part (b) $(-x)(-y) > 0$. Using part (e) in **Exercise 1** we have the equivalent expression $-[x(-y)] > 0$, using the same result we obtain the equivalent expression $-[-(xy)] > 0$ By part (d) in **Exercise 1** we obtain the equivalent expression $xy > 0$. Thus, we have shown that if x and y are both positive or both negative, then $xy > 0$.
Let $xy > 0$. Assume for contradiction $x > 0$ and $y < 0$. Then, by part (c) it must be that $-y > 0$. By axiom (6), $x(-y) > 0$. By part (e) in **Exercise 1** we obtain the equivalent expression $-(xy) > 0$. By part (c) we have that $xy < 0$. This is a contradiction since $xy > 0$. Therefore, it cannot be that $x > 0$ and $y < 0$. We can use this argument of contradiction with the case $x < 0$ and $y > 0$ to show that this is also not possible. If either x or y is zero, then their product is zero, which also contradicts the fact $xy > 0$. Thus, the only cases that remain is that x and y are both positive or both negative, which as proven earlier does not cause a contradiction. Therefore, if $xy > 0$, then it must be that x and y are both positive or both negative. □

(i) $x > 0 \Rightarrow 1/x > 0$.
Proof. Let $x > 0$. Note $x \cdot (1/x) = 1$ by axiom (4). Furthermore, by part (g) we have $1 > 0$. Thus, $x \cdot (1/x) > 0$. By part (h) this implies that x and $1/x$ are both positive or both negative. Since $x > 0$ it must be that $1/x > 0$. □

(j) $x > y > 0 \Rightarrow 1/x < 1/y$.
Proof. Let $x > y > 0$. By part (i) $1/x > 0$ and $1/y > 0$. Thus, by axiom (6) $x \cdot (1/x) > y \cdot (1/x)$, which is equivalent to $1 > y \cdot (1/x)$. Similarly, by axiom (6) we have $1 \cdot (1/y) > y \cdot (1/x) \cdot (1/y)$. This last expression, using axioms (1), (2), and (3), can be simplified into the expression $1/x < 1/y$. □

(k) $x < y \Rightarrow x < (x + y)/2 < y$.
Proof. Let $x < y$. By axiom (6) we have $x + y < y + y$. Using axiom (5) we obtain the equivalent expression $x+y < y(1+1) = y \cdot 2$. Since $1+1 = 2 > 0$, we note $1/2 > 0$ by part (i). Furthermore, by axiom (6) we obtain the expression $(x+y)(1/2) < y \cdot 2 \cdot (1/2) = y$. We can use these same steps to show $x < (x + y) \cdot (1/2)$, where the only change comes from the initial step, instead of adding y we add x. Therefore, we have $x < (x + y)/2 < y$. □

3.

(a) Show that if $\mathcal{A}$ is a collection of inductive sets, then the intersection of the elements of $\mathcal{A}$ is an inductive set.
Proof. Let $\mathcal{A}$ be a collection of inductive sets. Consider $\bigcap_{A \in \mathcal{A}} A$. By definition, if A is inductive, then A contains 1. Since $\mathcal{A}$ is a collection of inductive sets, it must be that if $A \in \mathcal{A}$ then $1 \in A$. Thus, we conclude $1 \in \bigcap_{A \in \mathcal{A}} A$. Let $a \in \bigcap_{A \in \mathcal{A}} A$, this implies $a \in A$ for all $A \in \mathcal{A}$. Since $A \in \mathcal{A}$ is inductive, and $a \in A$, it must be that $a + 1 \in A$ also. Since this is true for all A containing a, we conclude $a + 1 \in \bigcap_{A \in \mathcal{A}} A$. Therefore, $\bigcap_{A \in \mathcal{A}} A$ is an inductive set. □

(b) Prove the basic properties (1) and (2) of $\mathbb{Z}_+$.
Proof. The basic properties of $\mathbb{Z}_+$ are: (1) $\mathbb{Z}_+$ is inductive, (2) if A is an inductive set of positive integers, then $A = \mathbb{Z}_+$. For (1) we note that $\mathbb{Z}_+$ is defined as $\mathbb{Z}_+ = \bigcap_{A \in \mathcal{A}} A$ where $\mathcal{A}$ is the collection of all inductive sets in $\mathbb{R}$. By part (a) we conclude $\mathbb{Z}_+$ is an inductive set.
Let A be an inductive set of positive integers. Since A is inductive, we note $1 \in A$. Furthermore, given that A is a set of positive integers, if $a \in A$ then a is a positive integer. Thus, $a \in \mathbb{Z}_+$. Therefore, $A \subseteq \mathbb{Z}_+$. Now, consider the set $\mathbb{Z}_+$, which is an inductive set of positive integers. As such, $1 \in \mathbb{Z}_+$, and if $a \in \mathbb{Z}_+$ then a is a positive integer. Thus, $a \in A$ and $\mathbb{Z}_+ \subseteq A$. Therefore, we conclude $\mathbb{Z}_+ = A$. □

4.

(a) Prove by induction that given $n \in \mathbb{Z}_+$, every non-empty subset of $\{1, 2, \dots, n\}$ has a largest element.
Proof. If $n = 1$, then every non-empty subset of $\{1\}$ has a largest element, since it is a set of one element. Assume that for $n \in \mathbb{Z}_+$, every non-empty subset of $\{1, \dots, n\}$ has a largest element. Consider now $n + 1 \in \mathbb{Z}_+$ and the set $\{1, \cdots, n + 1\}$. The subset $\{n + 1\}$ is a set of one element, and its largest element is $n + 1$. Thus, consider a non-empty subset that is

not $\{n+1\}$, call it A. Note $A \cap \{1, \dots, n\}$ is a subset of $\{1, \dots, n\}$ and therefore has a largest element, say a. If $n+1 \in A$, we note $a < n+1$ and thus $n+1$ is the largest element of A. If $n+1 \notin A$, then a is the largest element of A. Thus, we've shown that every non-empty subset of $\{1, \dots, n+1\}$ has a largest element. Hence, by the induction principle, for $n \in \mathbb{Z}_+$ we have proven that every non-empty subset of $\{1, \dots, n\}$ has a largest element. □

(b) Explain why you cannot conclude from (a) that every non-empty subset of $\mathbb{Z}_+$ has a largest element.

Proof. In (a) we only considered finite non-empty subsets of $\mathbb{Z}_+$. In particular, note that the non-empty subset of $\mathbb{Z}_+$, $\mathbb{Z}_+$ itself, has no largest element. This is due to the fact $\mathbb{Z}_+$ is an inductive set and for any $a \in \mathbb{Z}_+$ we have $a+1 \in \mathbb{Z}_+$. □

5. Prove the following properties of $\mathbb{Z}$ and $\mathbb{Z}_+$:

(a) $a, b \in \mathbb{Z}_+ \Rightarrow a+b \in \mathbb{Z}_+$. [Hint: Show that given $a \in \mathbb{Z}_+$, the set $X = \{x \mid x \in \mathbb{R} \text{ and } a+x \in \mathbb{Z}_+\}$ is inductive.]

Proof. Let $a \in \mathbb{Z}_+$ and consider the set $X = \{x \mid x \in \mathbb{R} \text{ and } a+x \in \mathbb{Z}_+\}$. Note since $a \in \mathbb{Z}_+$ it must also be that $a+1 \in \mathbb{Z}_+$. Hence, $1 \in X$. Let $x \in X$, this implies $a+x \in \mathbb{Z}_+$. Consider $x+1$ and note $a+(x+1) = (a+x)+1$. Since $a+x \in \mathbb{Z}_+$, it must also be that $(a+x)+1 \in \mathbb{Z}_+$. Therefore, $x+1 \in X$. Thus, X is an inductive set. Since X is inductive, we note $\mathbb{Z}_+ \subseteq X$. Hence, if $b \in \mathbb{Z}_+$ then $b \in X$. Therefore, $a+b \in \mathbb{Z}_+$ when $a, b \in \mathbb{Z}_+$. □

(b) $a, b \in \mathbb{Z}_+ \Rightarrow a \cdot b \in \mathbb{Z}_+$.

Proof. Let $a, b \in \mathbb{Z}_+$. By definition, $a \cdot b = a + a + \dots + a$ where we add b many times a to itself. If $b = 1$, we have $a \cdot 1 = a \in \mathbb{Z}_+$. Assume that $a \cdot n \in \mathbb{Z}_+$ for some $n \in \mathbb{Z}_+$. Consider $n+1$, and note $a \cdot (n+1) = (a \cdot n) + a$. Since $a, na \in \mathbb{Z}_+$, we have by part (a) that $a + na \in \mathbb{Z}_+$. Therefore, $(n+1)a \in \mathbb{Z}_+$. Hence, by the principle of induction we have shown that if $a, b \in \mathbb{Z}_+$ then $a \cdot b \in \mathbb{Z}_+$. □

(c) Show that $a \in \mathbb{Z}_+ \Rightarrow a-1 \in \mathbb{Z}_+ \cup \{0\}$. [Hint: Let $X = \{x \mid x \in \mathbb{R} \text{ and } x-1 \in \mathbb{Z}_+ \cup \{0\}\}$; show that X is inductive].

Proof. Let $a \in \mathbb{Z}_+$. Define $X = \{x \mid x \in \mathbb{R} \text{ and } x-1 \in \mathbb{Z}_+ \cup \{0\}\}$. If $x = 1$, note $x-1 = 0$ and thus $x \in \mathbb{Z}_+ \cup \{0\}$. Let $x \neq 1$ be an element of X. Consider $x+1$. Note $(x+1)-1 = x$. Since $x \in X$, $x-1 \in \mathbb{Z}_+ \cup \{0\}$. Given that $x \neq 1$, we conclude $x-1 \in \mathbb{Z}_+$. Since $\mathbb{Z}_+$ is an inductive set, $x \in \mathbb{Z}_+$. Therefore, $x \in \mathbb{Z}_+ \cup \{0\}$ which implies $x+1 \in X$. Hence, X is an inductive set. Since X is an inductive set, we note $\mathbb{Z}_+ \subseteq X$. Therefore, if $a \in \mathbb{Z}_+$, it must be $a \in X$. Thus, for $a \in \mathbb{Z}_+$ we conclude $a-1 \in \mathbb{Z}_+ \cup \{0\}$. □

(d) $c, d \in \mathbb{Z} \Rightarrow c+d \in \mathbb{Z}$ and $c-d \in \mathbb{Z}$. [Hint: Prove it first for $d = 1$.]

Proof. Let $c \in \mathbb{Z}$. This implies either $c = 0$, $c \in \mathbb{Z}_+$ or $-c \in \mathbb{Z}_+$. Consider then $c+1$. If $c = 0$, then $c+1 = 1 \in \mathbb{Z}$ and $c-1 = -1 \in \mathbb{Z}$. If $c \in \mathbb{Z}_+$, by part (c) $c-1 \in \mathbb{Z}_+ \cup \{0\} \subset \mathbb{Z}$. Furthermore, by the fact $\mathbb{Z}_+$ is an inductive set, $c+1 \in \mathbb{Z}_+ \subset \mathbb{Z}$. Finally, if $-c \in \mathbb{Z}_+$, consider $c-1$, and note $-(c-1) = (-c)+1 \in \mathbb{Z}_+$ since $\mathbb{Z}_+$ is an inductive set. Similarly, consider $c+1$, and note $-(c+1) = -c-1 \in \mathbb{Z}_+ \cup \{0\} \subset \mathbb{Z}$ by part (c). This implies that both $c+1$ and $c-1$ are elements of $\mathbb{Z}$. Therefore, we conclude that if $c \in \mathbb{Z}$, then $c+1$ and $c-1$ are also in $\mathbb{Z}$.

Let $c, d \in \mathbb{Z}$. If $d = 0$ we note $c + d = c \in \mathbb{Z}$. Thus, assume $d \in \mathbb{Z}_+$. We already showed that if $d = 1$ then $c + d \in \mathbb{Z}$. Assume that if $d = n \in \mathbb{Z}_+$ then $c + d \in \mathbb{Z}$. We show that $c + (n + 1)$ is also in $\mathbb{Z}$. Simply note $c + (n + 1) = (c + n) + 1$, by assumption $c + n \in \mathbb{Z}$ and by what we already showed we have that $(c + n) + 1 \in \mathbb{Z}$. Therefore, if $d \in \mathbb{Z}_+$ then $c + d \in \mathbb{Z}$. Now, consider $c - d$. We note that if $d = 1$ we already showed that $c - d \in \mathbb{Z}$. Thus, assume that $c - n \in \mathbb{Z}$ for some $n \in \mathbb{Z}_+$. Consider now $n + 1$ and note $c - (n + 1) = (c - n) - 1$. From assumption, $c - n \in \mathbb{Z}$ and from what we showed earlier we also have $(c - n) - 1 \in \mathbb{Z}$. Therefore, if $d \in \mathbb{Z}_+$ we have proved $c + d$ and $c - d$ are both elements of $\mathbb{Z}$.

Let $c, d \in \mathbb{Z}$. Assume $-d \in \mathbb{Z}_+$. Consider then $c + d$, if $d = -1$ we note $c - 1 \in \mathbb{Z}$. Assume then that $c + n \in \mathbb{Z}$ for some n such that $-n \in \mathbb{Z}_+$. Then note $c + (n + 1) = (c + n) + 1$. From assumption $c + n \in \mathbb{Z}$ and since we are adding 1 we further conclude $(c + n) + 1 \in \mathbb{Z}$ also. Thus, we have shown that $c + d \in \mathbb{Z}$ if $-d \in \mathbb{Z}_+$. Now consider $c - d$. If $d = -1$, we note $c - d = c + 1 \in \mathbb{Z}$. Assume then that $c - n \in \mathbb{Z}$ for some n such that $-n \in \mathbb{Z}_+$. Note then $c - (n + 1) = c + (-n - 1) = (c - n) - 1$. By assumption, $c - n \in \mathbb{Z}$, and thus we conclude $(c - n) - 1 \in \mathbb{Z}$. Therefore, we have shown that if $-d \in \mathbb{Z}$, then $c + d$ and $c - d$ are elements of $\mathbb{Z}$.

Therefore, we have proved that for $c, d \in \mathbb{Z}$, then $c + d, c - d \in \mathbb{Z}$. □

(e) $c, d \in \mathbb{Z} \Rightarrow c \cdot d \in \mathbb{Z}$.

Proof. Let $c, d \in \mathbb{Z}$. Note $c \cdot d = c + c + ... + c$ where we add c to itself d many times. If $d = 1$, then $c \cdot 1 = c \in \mathbb{Z}$. Furthermore, if $d = 0$ then $c \cdot 0 = 0 \in \mathbb{Z}$. Thus, consider $d \in \mathbb{Z}_+$. Assume that if $d = n \in \mathbb{Z}_+$ then $c \cdot d \in \mathbb{Z}$. Consider then $d = n + 1$ and note $c \cdot d = c(n + 1) = cn + c$. Since $cn \in \mathbb{Z}$ by assumption, by part (d) $cn + c \in \mathbb{Z}$. Therefore, if $d \in \mathbb{Z}_+$ we conclude that $c \cdot d \in \mathbb{Z}$. Note now if $d = -1$, then $c \cdot d = -c$. It must be that $-c \in \mathbb{Z}$ (since it is a consequence no matter if $c = 0$, $c \in \mathbb{Z}_+$ or $-c \in \mathbb{Z}_+$). Now, let $d = -n$ where $n \in \mathbb{Z}_+$ and assume $c \cdot d \in \mathbb{Z}$. Now consider the case $d = -(n + 1)$. Then, $c \cdot d = c \cdot (-n - 1) = -cn - c$. By assumption, $-cn \in \mathbb{Z}$ and thus by part (d) we conclude $-cn - c \in \mathbb{Z}$. Hence, we have proven that if $c, d \in \mathbb{Z}$, then $c \cdot d \in \mathbb{Z}$. □

6. Let $a \in \mathbb{R}$. Define inductively

$$a^1 = a,$$

$$a^{n+1} = a^n \cdot a$$

for $n \in \mathbb{Z}_+$. Show that for $n, m \in \mathbb{Z}_+$ and $a, b \in \mathbb{R}$,

$$a^n a^m = a^{n+m}, \quad (a^n)^m = a^{nm}, \quad a^m b^m = (ab)^m.$$

These are called ***the laws of exponents***, [Hint: For fixed n, prove the formulas by induction on m.]

Proof. Let $a \in \mathbb{R}$ and $n, m \in \mathbb{Z}_+$. To prove the first identity, $a^n a^m = a^{n+m}$, first assume $m = 1$. Note $a^n a^m = a^n \cdot a$. By definition, $a^{n+1} = a^n \cdot a$, thus the identity holds when $m = 1$. Assume that the identity holds for $m = k$, consider now the case $m = k + 1$. Note the product $a^n \cdot a^{k+1}$ is equivalent to $a^n \cdot (a^k \cdot a)$. Furthermore, $a^n \cdot (a^k \cdot a) = (a^n \cdot a^k) \cdot a$. From assumption,

$(a^n \cdot a^k) \cdot a = a^{n+k} \cdot a$, and by definition, $a^{n+k} \cdot a = a^{n+(k+1)}$. Therefore, the identity $a^n \cdot a^m = a^{n+m}$ holds by induction.

To prove the second identity, $(a^n)^m = a^{nm}$, first let $m = 1$. Note $(a^n)^1 = a^n$ by definition, and thus the identity holds when $m = 1$. Assume now the identity holds for $m = k$, and consider the case $m = k + 1$. Then, $(a^n)^m = (a^n)^{k+1}$, which by definition is equivalent to $(a^n)^k \cdot a^n$. From assumption, note the expression is equivalent to $a^{nk} \cdot a^n$. Using the identity proven already, note $a^{nk} \cdot a^n = a^{nk+n} = a^{n(k+1)}$. Hence, the identity $(a^n)^m = a^{nm}$ hold by induction.

Finally, to prove the third identity, $a^m b^m = (ab)^m$, consider the case $m = 1$. Note $a^m b^m = ab = (ab)^1$ by assumption and definition. Thus, the identity holds when $m = 1$. Assume now the identity holds for $m = k$, and consider now the case $m = k + 1$. Then, the product $a^{k+1} b^{k+1} = a^k \cdot a \cdot b^k \cdot b$ by the first identity. Reordering, $a^k \cdot a \cdot b^k \cdot b = (a^k \cdot b^k) \cdot ab$, and thus by assumption $(a^k \cdot b^k) \cdot ab = (ab)^k \cdot ab$, which by the first identity is equivalent to $(ab)^{k+1}$. Therefore, the third identity holds by induction. □

7. Let $a \in \mathbb{R}$ and $a \neq 0$. Define $a^0 = 1$, and for $n \in \mathbb{Z}_+$, $a^{-n} = 1/a^n$. Show that the laws of exponents hold for $a, b \neq 0$ and $n, m \in \mathbb{Z}$.

Proof. Let $a, b \in \mathbb{R}$ such that $a, b \neq 0$, and let $n, m \in \mathbb{Z}$. By Exercise 6. the identities hold when $n, m \in \mathbb{Z}_+$, and thus it need only be shown that they also hold when $-n, -m$ are elements of $\mathbb{Z}_+ \cup \{0\}$.

As such, let $-n, -m \in \mathbb{Z}_+ \cup \{0\}$. To prove the identity $a^n a^m = a^{n+m}$ holds first assume $m = 0$. Note $a^n a^m = a^n \cdot 1 = a^n$ by definition, and thus the identity holds when $m = 0$. If $m = -1$, note $a^n a^m = a^n a^{-1}$. By definition, $a^n a^{-1} = a^n(1/a)$ and since a^n is defined as $a^{n-1} \cdot a$, we conclude $a^n a^{-1} = a^{n-1} \cdot (a \cdot 1/a) = a^{n-1}$. Hence, the identity also holds when $m = -1$. Assume $m = k$ for some k such that $-k \in \mathbb{Z}_+$, consider then the case $m = k - 1$. Note $a^n a^{k-1} = a^n(1/a^{-k+1})$ by definition, furthermore, $a^n a^{k-1} = a^n \cdot (1/[a^{-k}a]) = (a^n \cdot 1/a^{-k})(1/a)$ by both the identities proven in **6.** and the results from **1.**. By definition, $(a^n \cdot 1/a^{-k}) \cdot (1/a) = (a^n a^k)a^{-1}$, and by assumption, $(a^n a^k)a^{-1} = a^{n+k}a^{-1}$. Since the identity holds when $m = -1$, $a^{n+1}a^{-1} = a^{n+(k-1)}$. Thus, the identity $a^n a^m = a^{nm}$ holds when $-n, -m \in \mathbb{Z}_+ \cup \{0\}$ by induction.

To prove the second identity, $(a^n)^m = a^{nm}$ holds when $-n, -m \in \mathbb{Z}_+ \cup \{0\}$ consider the case $m = 0$. Note then $(a^n)^m = (a^n)^0$, which by definition is equivalent to 1, and thus the identity holds when $m = 0$. Now assume $m = -1$, and note that by definition $(a^n)^{-1}$ is equivalent to $1/(a^n)^1$. Furthermore, by definition $1/(a^n)^1 = 1/(a^n) = a^{-n}$, and thus the identity holds when $m = -1$. Assume then that $m = k$ for some k such that $-k \in \mathbb{Z}_+$. Consider the case $m = k - 1$, and note $(a^n)^m = (a^n)^{k-1}$. Using the first identity we proved, note $(a^n)^{k-1} = (a^n)^k \cdot (a^n)^{-1}$. By assumption, $(a^n)^k = a^{nk}$ and since the identity holds when $m = -1$, note $(a^n)^k \cdot (a^n)^{-1} = a^{nk} \cdot a^{-n}$. Finally, using the first identity once more, $a^{nk}a^{-n} = a^{nk-n} = a^{n(k-1)}$. Therefore, the second identity holds even when $-n, -m \in \mathbb{Z}_+ \cup \{0\}$.

Finally, to prove the last identity, $a^m b^m = (ab)^m$, first let $m = 0$. Note $a^0 b^0 = 1 \cdot 1 = 1$, and thus the identity holds when $m = 0$. Now consider the case $m = -1$, and note $a^m b^m = a^{-1}b^{-1} = (1/a)(1/b) = 1/(ab) = (ab)^{-1}$. Thus, the identity also holds when $m = -1$. Assume

then that the identity holds when $m = k$ for some k such that $-k \in \mathbb{Z}_+$. Then, the product $a^{k-1}b^{k-1} = (a^k a^{-1})(b^k b^{-1})$ by the first identity. Reordering the expression, $(a^k a^{-1})(b^k b^{-1}) = (a^k b^k)(a^{-1}b^{-1})$, which is equivalent to $(ab)^k(a^{-1}b^{-1})$ by assumption. Furthermore, since the identity holds when $m = -1$, $(ab)^k(a^{-1}b^{-1}) = (ab)^k(ab)^{-1}$. By the first identity, $(ab)^k(ab)^{-1} = (ab)^{k-1}$. Therefore, the third identity also holds when $-n, -m \in \mathbb{Z}_+ \cup \{0\}$.

As such, it has been shown that the identities from exercise **6.** hold even when n, m are allowed to be elements of $\mathbb{Z}$. □

8.

(a) Show that $\mathbb{R}$ has the greatest lower bound property.
Proof. From axiom (7) the order relation $<$ has the least upper bound property on $\mathbb{R}$. By Exercise **13** in the Order Relations solutions $<$ also has the greatest lower bound property on $\mathbb{R}$. □

(b) Show that $\inf\{1/n \mid n \in \mathbb{Z}_+\} = 0$.
Proof. To show $\inf\{1/n \mid n \in \mathbb{Z}_+\} = 0$ first note that it is a non-empty subset of $\mathbb{R}$, and that is is bounded below thus it has a greatest lower bound. As such, it need only be shown that 0 is a lower bound of the set, and that it is the greatest lower bound as well. By construction, note $1/n > 0$ since $n \in \mathbb{Z}_+$. Let a be a lower bound of the set, then $a \leq 1/n$ for all $1/n$ in the set. Note if a is an element of the set, then $a = 1/m$ for some $m \in \mathbb{Z}_+$. Since $m + 1 \in \mathbb{Z}_+$ given that $\mathbb{Z}_+$ is inductive, the element $1/(m+1)$ is also in the set. Since $m < m + 1$ note $1/m > 1/(m+1)$, which implies that a is not the smallest element of the set and therefore not a lower bound for the set. Hence, a is not an element of the set and the set has no smallest element. Assume for contradiction $a > 0$, then note $a < 1$ since 1 is in the set. As such, $0 < a < 1$. Furthermore, since $a > 0$ and $1/n > 0$ for all $n \in \mathbb{Z}_+$, this implies $1/a > n$ for all $n \in \mathbb{Z}_+$. This implies $\mathbb{Z}_+$ is a bounded above subset of $\mathbb{R}$ and thus has a least upper bound. Let b be the least upper bound of $\mathbb{Z}_+$, then $b - 1 < n$ for some $n \in \mathbb{Z}_+$. Note then this implies $n + 1 > b$ which is a contradiction to b being an upper bound of $\mathbb{Z}_+$. Therefore, $\mathbb{Z}_+$ has no least upper bound, implying it is not bounded above. Therefore, $1/a > n$ for all $n \in \mathbb{Z}_+$ is not true, implying $a > 0$ is not true. As such, for a a lower bound of A it must be that $a < 0$. Hence, we conclude 0 is the greatest lower bound of the set. □

(c) Show that given a with $0 < a < 1$, $\inf\{a^n \mid n \in \mathbb{Z}_+\} = 0$. [Hint: Let $h = (1-a)/a$, and show that $(a+h)^n \geq 1 + nh$.]
Proof. Let $a \in \mathbb{R}$ such that $0 < a < 1$ and define $A = \{a^n \mid n \in \mathbb{Z}_+\}$. Note A is a non-empty subset of $\mathbb{R}$, given that $a \in A$. Furthermore, A is bounded below since $0 < a^n$ for any $n \in \mathbb{Z}_+$. Therefore, A has a greatest lower bound in $\mathbb{R}$, call it α. To prove that there exists no positive lower bound on A, it will be shown that for $0 < a < 1$ and $x > 0$, $a^n < x$ for some $n \in \mathbb{Z}_+$. Since $0 < a < 1$ note $1/a > 1$. Thus, $1/a - 1 = (1-a)/a > 0$. Define $h = (1-a)/a$, and note $1 + h = 1/a$ by construction.
Using the hint, it will be proven by induction that $(1+h)^n \geq 1 + nh$ for $n \in \mathbb{Z}_+$. Thus, note if $n = 1$ then the identity holds since $1 + h \geq 1 + h$. Now, assume for induction that the identity holds for some $k \in \mathbb{Z}_+$, and thus consider $k + 1$. In particular,

$(1+h)^{k+1} = (1+h)^k(1+h) \geq (1+kh)(1+h)$ by the induction hypothesis. Expanding, note $(1+kh)(1+h) = 1+h+kh+kh^2$. Since $h, k \geq 0$, $1+h+kh+kh^2 \geq 1+h+kh = 1+(k+1)h$. Therefore, $(1+h)^{k+1} \geq 1+(k+1)h$ and thus the identity holds for $n \in \mathbb{Z}_+$.
Consider the real number $[(1/x)-1]/h$, and note there exists some $n \in \mathbb{Z}_+$ such that $n > [(1/x)-1]/h$. This implies $nh > (1/x)-1$, and furthermore that $1+nh > 1/x$. By the identity proven, $(1+h)^n > 1/x$. Using the definition of h, $(1/a)^n > 1/x$. This implies $a^n < x$. Therefore, it has been shown that for any $x > 0$ there exists some $n \in \mathbb{Z}_+$ such that $a^n < x$ for $0 < a < 1$. Hence, if x is a lower bound of A, it cannot be that $x > 0$. Therefore, $\alpha = 0$. □

9.

(a) Show that every non-empty subset of $\mathbb{Z}$ that is bounded above has a largest element.
Proof. Let $A \subset \mathbb{Z}$ be a non-empty and bounded above subset of $\mathbb{Z}$. Since $\mathbb{Z}$ is a subset of $\mathbb{R}$, $A \subset \mathbb{R}$. Furthermore, given that $\mathbb{R}$ has the least upper bound property, A must have a least upper bound in $\mathbb{R}$. Call that least upper bound α. Thus, for any $a \in A$, $a \leq \alpha$. If $\alpha \notin A$, then for any $a_0 \in A$ there exists some $a_1 \in A$ such that $a_1 > a_0$. Otherwise, if no such a_1 exists it implies $a_0 \geq a$ for all $a \in A$, contradicting the fact α is the least upper bound of A. Note since $a_1 > a_0$ it must be $a_1 + n = a_0$ for some $n \in \mathbb{Z}_+$. In particular, since this is true for any $a \in A$, $a_1 + 1 \in A$ also. This contradicts the fact A is bounded above. Therefore, it must be $\alpha \in A$ and thus A has a largest element. □

(b) If $x \notin \mathbb{Z}$, show there is exactly one $n \in \mathbb{Z}$ such that $n < x < n+1$.
Proof. Let $A \subset \mathbb{Z}$ be the set of all integers smaller than $x \notin \mathbb{Z}$. Since A is bounded above and is a subset of $\mathbb{Z}$ (a) guarantees that A has a largest element. Let n be the largest element of A. Since n is the largest element of A, it must be $n+1 \notin A$, and thus $n+1 > x$. Given that $n \in A$, $n < x$, and therefore $n < x < n+1$. □

(c) If $x - y > 1$, show there is at least one $n \in \mathbb{Z}$ such that $y < n < x$.
Proof. Let $x, y \in \mathbb{R}$ such that $x-y > 1$. This implies $y+1 < x$, note also that $y < y+1 < x$. Since $y+1$ is an element of $\mathbb{R}$, by (b) there exists some $n \in \mathbb{Z}$ such that $n < y+1 < n+1$. Furthermore, $n-1 < y < n$, and thus $y < n < y+1 < x$. Therefore, $y < n < x$ for some $n \in \mathbb{Z}$. □

(d) If $y < x$, show there is a rational number z such that $y < z < x$.
Proof. Let $x, y \in \mathbb{R}$ such that $y < x$. If $x - y > 1$ by part (c) note there exists some $m \in \mathbb{Z}$ such that $y < m < x$. Thus, assume $x - y \leq 1$. In particular, note $0 < x - y \leq 1$. By Exercise 8 part (b) there exists some $n \in \mathbb{Z}_+$ such that $0 < 1/n < x - y$. Since $n > 0$, $n(1/n) < n(x-y)$ which is equivalent to $1 < n(x-y)$. This implies, $nx - ny > 1$ with $ny < nx$. By part (c) there exists some $t \in \mathbb{Z}$ such that $ny < t < nx$. Given that $n > 0$, it must be $y < t/n < x$ for t/n a rational number. □

10. Show that every positive number a has exactly one positive square root, as follows:

(a) Show that if $x > 0$ and $0 \leq h < 1$, then

$$(x+h)^2 \leq x^2 + h(2x+1),$$

$$(x-h)^2 \geq x^2 - h(2x).$$

Proof. For the second identity, note $(x-h)^2 = x^2 - 2xh + h^2$ and $x^2 - h(2x) = x^2 - 2xh$. Since $0 \leq h < 1$, $h^2 \geq 0$, and thus $(x-h)^2 \geq x^2 - h(2x)$.
For the first identity, note $(x+h)^2 = x^2 + 2xh + h^2$ and $x^2 + h(2x+1) = x^2 + 2xh + h$. Since $0 \leq h < 1$, note $0 \leq h^2 < h$. Therefore, $(x+h)^2 \leq x^2 + h(2x+1)$. □

(b) Let $x > 0$. Show that if $x^2 < a$, then $(x+h)^2 < a$ for some $h > 0$; and if $x^2 > a$, then $(x-h)^2 > a$ for some $h > 0$.

Proof. Let $x > 0$ such that $x^2 < a$. Then by part (a), $(x+h)^2 \leq x^2 + h(2x+1)$ for $0 \leq h < 1$. Let $\epsilon = a - x^2$, and note

$$0 < \frac{\epsilon}{2x+1+\epsilon} < 1.$$

Let h be the above fraction, and note $h(2x+1) < \epsilon$ since

$$\frac{2x+1}{2x+1+\epsilon} \cdot \epsilon < \epsilon.$$

Therefore, $(x+h)^2 < x^2 + \epsilon = x^2 + a - x^2 = a$ for some $h > 0$.
Let $x > 0$ such that $x^2 > a$. By part (a), $(x-h)^2 \geq x^2 - h(2x)$ for $0 \leq h < 1$. Define $\epsilon = x^2 - a$ and note

$$0 < \frac{\epsilon}{2x+\epsilon} < 1.$$

Let h be the above fraction, and note $h(2x) < \epsilon$ since

$$\frac{2x}{2x+\epsilon} \cdot \epsilon < \epsilon.$$

This implies $x^2 - h(2x) > x^2 - \epsilon = x^2 - x^2 + a = a$. Therefore, $(x-h)^2 > a$ for some $h > 0$. □

(c) Given $a > 0$, let B be the set of all real numbers x such that $x^2 < a$. Show that B is bounded above and contains at least one positive number. Let $b = \sup B$; show that $b^2 = a$.

Proof. Let $a > 0$ and B be the set of all real numbers x such that $x^2 < a$. Note B is bounded above sine for $x \in \mathbb{R}$, either $x \leq x^2$ or $x > x^2$. If $x \leq x^2$ then $x < a$ and if $x > x^2$ note it must be $x < 1$. Thus, if $a \geq 1$ then a is an upper bound of B, and if $1 > a$ then 1 is an upper bound of B. In either case, there exists an upper bound for B and thus B is bounded above. Since $a > 0$, $0 \in B$. By part (b) there exists some $h > 0$ such that $(0+h)^2 < a$ which implies there exists some $h > 0$ such that $h^2 < a$. Therefore, there exists some positive real number h that is an element of B.
Since $B \subset \mathbb{R}$ is bounded above it has a least upper bound, call it b. Since B contains at least one positive real number, $b > 0$. Assume for contradiction $b^2 < a$. Then, by part by part (b) $(b+h)^2 < a$ for some $h > 0$. Thus, $b + h \in B$. This is a contradiction since $b + h > b$ and $b + h \in B$ cannot both be true given b is the least upper bound of B. Thus, assume for contradiction $b^2 > a$. Then by part (b) there exists some $h > 0$ such that $(b-h)^2 > a$. Thus, $b - h$ is an upper bound of B while also $b - h < b$ which is a contradiction given that b is the least upper bound of B. Therefore, it must be that $b^2 = a$. □

(d) Show that if b and c are positive and $b^2 = c^2$, then $b = c$.

Proof. Let $b, c \in \mathbb{R}$ such that $b, c > 0$ and $b^2 = c^2$. Therefore, $b^2 - c^2 = 0$. Note $b^2 - c^2 = (b + c)(b - c)$. Thus,

$$(b + c)(b - c) = 0.$$

This implies either $b + c = 0$ or $b - c = 0$. Since $b, c > 0$ it must be $b + c > 0$. Hence, $b - c = 0$ implying $b = c$. □

11. Given $m \in \mathbb{Z}$, we say that m is ***even*** if $m/2 \in \mathbb{Z}$, and m is ***odd*** otherwise.

(a) Show that if m is odd, $m = 2n+1$ for some $n \in \mathbb{Z}$. [Hint: Choose n so that $n < m/2 < n+1$.]
Proof. Let $m \in \mathbb{Z}$ such that m is odd. Since m is odd, $m/2 \notin \mathbb{Z}$. Thus, by Exercise 9 part (b) there exists some $n \in \mathbb{Z}$ such that $n < m/2 < n + 1$. This is equivalent to $2n < m < 2n + 2$. Since $m \in \mathbb{Z}$, it must be that $m = 2n + 1$. □

(b) Show that if p and q are odd, so are $p \cdot q$ and p^n, for any $n \in \mathbb{Z}_+$.
Proof. Let $p, q \in \mathbb{Z}$ be odd. By part (a) $p = 2n+1$ and $q = 2m+1$ for some $n, m \in \mathbb{Z}$. Thus, $p \cdot q = (2n+1)(2m+1) = 4nm+2n+2m+1 = 2(2nm+n+m)+1$. Since $2nm+n+m \in \mathbb{Z}$, it can be concluded $p \cdot q$ is odd since $p \cdot q = 2t + 1$ for some $t \in \mathbb{Z}$ (which makes evident $(pq)/2 \notin \mathbb{Z}$).
Let $p \in \mathbb{Z}$ be odd. Note $p^1 = p$ which is odd. Assume for induction that p^k is odd for some $k \in \mathbb{Z}_+$. Consider then $k + 1$, and note $p^{k+1} = p^k \cdot p$. By assumption p^k is odd and so is p. By what was proven earlier it must be that $p^k \cdot p$ is odd also. Therefore, p^{k+1} is odd and it can be concluded from induction that p^n is odd for any $n \in \mathbb{Z}_+$ given that p is odd. □

(c) Show that if $a > 0$ is rational, then $a = m/n$ for some $m, n \in \mathbb{Z}_+$ where not both m and n are even. [Hint: Let n be the smallest element of the set $\{x \mid x \in \mathbb{Z}_+ \text{ and } x \cdot a \in \mathbb{Z}_+\}$.]
Proof. Let $a > 0$ be a rational number. Then, $a = m/n$ for some $m, n \in \mathbb{Z}$. Since $a > 0$ it cannot be that $m = 0$. Furthermore, m and n must be either both positive or both negative, since otherwise $a < 0$. In either case, m/n is equivalent to $(-m)/(-n)$. As such, $a = m/n$ for some $m, n \in \mathbb{Z}_+$. Let $S = \{x \in \mathbb{Z}_+ \mid x \cdot a \in \mathbb{Z}_+\}$, and note it is non-empty since $na = m$ by construction. In particular, by the Well-Ordering property S has a smallest element. Let n be the smallest element of S, thus $an = m$. Assume for contradiction both n and m are even. Then, $n = 2s$ and $m = 2t$ for some $s, t \in \mathbb{Z}_+$ (recall we chose $n, m \in \mathbb{Z}_+$). Hence, $na = m$ is equivalent to $2s \cdot a = 2t$. Note this implies $sa = t$ where $s < 2s = n$ and $t \in \mathbb{Z}_+$. Therefore, $s \in S$ which is a contradiction since $s < n$. As such, both n and m cannot be event.
Hence, it has been proven that if a is a positive rational number, then $a = m/n$ for some $n, m \in \mathbb{Z}_+$ where not both m and n are even. □

(d) *Theorem. $\sqrt{2}$ is irrational.*
Proof. Assume for contradiction $\sqrt{2}$ is rational. Since $\sqrt{2} > 0$ by part (c) there exists $m, n \in \mathbb{Z}_+$ that are not both even such that $\sqrt{2} = m/n$. This implies $(m/n)^2 = 2$, which is equivalent to $m^2/n^2 = 2$. Thus, $m^2 = 2n^2$ implies m^2 is even. By part (b) it can be concluded m^2 being even implies m is even. Thus, $m = 2s$ for some $s \in \mathbb{Z}$. In particular, $m^2 = 4s^2 = 2n^2$. Hence, $n^2 = 2s^2$. Since $s^2 \in \mathbb{Z}_+$ it must be n^2 is even. Therefore, n is even

also. This is a contradiction since both m and n cannot be even. Therefore, $\sqrt{2} \neq m/n$, and thus $\sqrt{2}$ is not a rational number. $\square$